AJAN MUTHURAJ

Robotics Software & Autonomy | IIT Madras | Portfolio | GitHub | LinkedIn | ajanm2003@gmail.com

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

PROGRAM	INSTITUTION	CGPA / %	YEAR
Mechanical Engineering (B.Tech)	Indian Institute of Technology Madras	**8.52/10**	**2026**
Class XII (CBSE)	Bharatiya Vidya Bhavan's V. M. Public School	**95.4%**	**2021**
Class X (CBSE)	Bharatiya Vidya Bhavan's V. M. Public School	**97.2%**	**2019**

- Built a ROS 2 autonomy stack for indoor UAV exploration, integrating MAVROS, ArduPilot SITL and 3D SLAM in a nine-drone Isaac Sim demo.
- Placed second among Indian teams at the 2025-26 Swarm Rescue Challenge with an autonomous search-and-rescue solution for ten UAVs.
- Co-authored work on autonomous multi-UAV exploration presented as a poster at the XPLORE Workshop, IEEE ICRA 2026, Vienna, Austria.
- Integrated a UR5 arm and eBot for autonomous package sorting and delivery at e-Yantra, combining robot perception, planning and control.

RELEVANT SKILLS

Robotics: ROS 2, SLAM, Nav2, MoveIt 2, motion planning, RGB-D perception, state estimation, PID control, simulation and reinforcement learning.
Software and Tools: Python, C++, Linux, Git, Docker Compose, OpenCV, PyTorch, MATLAB, Gazebo, Isaac Sim, ArduPilot, SolidWorks and Arduino/ESP32.
Languages: English, Hindi, Tamil and basic Japanese. Interests include robot decision-making, humanoids, swarms and vision-language-action models.

PROFESSIONAL EXPERIENCE

Systems Engineer **DLTX, Japan** **Oct 2026 - Present**	• Practiced reverse and forward engineering on Lotus Notes applications to understand how legacy screens and data were represented. • Built a React interface from supplied XML exports by interpreting the source structure and translating it into web components. • Studied full-stack development across React, Java/Spring Boot, data persistence, Docker containers and Kubernetes fundamentals.
Humanoid Navigation and Manipulation **IIT Roorkee · Intern** **Jun - Jul 2025**	• Integrated Nav2, SLAM and differential-drive control for autonomous humanoid navigation in a SolidWorks-derived Gazebo simulation. • Connected RGB-D object detection and 3D localization to MoveIt 2 and OctoMap for collision-aware approach and grasp planning. • Reported about 96.6% grasp success over 30 simulation trials and recorded separate demonstrations of navigation and manipulation.

PROJECTS

EXTRACURRICULAR PROJECTS

Isaac Sim UAV Autonomy	• Integrated MAVROS and ArduPilot SITL with ROS 2 Jazzy, sensor topics and TF frames for RGB, IMU, range and LiDAR data. • Built collision-aware 3D frontier exploration using RTAB-Map odometry and OctoMap, with replanning, return-to-home and landing. • Published Docker Compose setup, launch instructions, RViz telemetry and the recording of a nine-drone simulation. • Added single-drone trajectory and SLAM benchmarks at 0.25, 0.5 and 1.0 m/s and an automated PID tuning workflow.
Time-Varying Reach-Avoid RL **B.Tech Thesis**	• Extended the discounted reach-avoid Bellman formulation with time as a state variable and built custom moving-goal environments. • Trained a Double DQN policy with experience replay and target-network updates for navigation under time-varying safety constraints. • Measured 94.4% predicted-safe precision in the two-obstacle simulation and published the thesis poster and evaluation figures.
Swarm Rescue Challenge 2026	• Contributed to adaptive frontier exploration and temporal occupancy-grid fusion for collaborative coverage by ten simulated drones. • Worked on FastSLAM/ICP localization and integrated hazard detection and collision recovery into the exploration workflow. • Placed second among Indian teams with IIT Madras and co-authored related work presented at the ICRA 2026 XPLORE Workshop.
Logistic coBot **e-Yantra, IIT Bombay**	• Integrated a UR5 arm and eBot rover with ROS 2 for package pickup, sorting and delivery in a logistics competition setting. • Used MoveIt 2 Servo and ArUco pose estimates for arm control, plus EKF state estimation and ultrasonic docking for the rover. • Validated package handling, rover localization and docking in Gazebo and on physical UR5 and eBot hardware at IIT Bombay.
Cubitect **Electronics Club** **Project Lead**	• Led a modular robotics project exploring self-reconfiguring cubes that use reaction wheels and controlled toppling to move. • Built vision-based tracking with A* path planning and Hungarian assignment to match individual cubes with target positions. • Developed the actuation and coordination approach and worked on a ROS 2 and Gazebo environment to test multi-cube behavior.
Swarm Navigation **Inter-IIT Tech Meet**	• Developed a ROS 2 and Gazebo simulation workflow for groups of mobile robots navigating a dynamic warehouse-like environment. • Worked across LiDAR/DBSCAN obstacle clustering, map merging and Twin Delayed DDPG training for navigation policies. • Published recordings, checkpoints and four- and five-robot launch configurations with the IIT Madras Inter-IIT robotics team.
GeoGuide **e-Yantra, IIT Bombay**	• Built a physical mobile robot that used an overhead camera to localize arena events and navigate to them in priority order. • Combined ResNet-50 and OpenCV classification with Dijkstra routing, ArUco tracking and QGIS georeferencing for the arena. • Sent route commands to an ESP32 rover over Wi-Fi/TCP and used infrared sensors for on-robot line following during execution.

ACADEMIC PROJECTS

Collective Swarm Navigation **ME5253, IIT Madras**	• Extended Olcay et al.'s decentralized flocking baseline in MATLAB with shared map memory for navigation in unknown environments. • Added occupancy-grid mapping, Deadlock Detection and Recovery (DDR), and Delayed Self-Reinforcement (DSR) for noisy sensing. • Co-authored with Aadityanshu Abhinav and compared goal convergence and exploration coverage against the baseline in simulation.
Adaptive RRT* Motion Planning	• Compared Adaptive RRT*, RRT*, A*, Adaptive A* and Lifelong Planning A* in simulated environments with dynamic obstacles. • Retained feasible route segments and replanned from a safe waypoint when a moving obstacle blocked the original path. • Evaluated planners on path efficiency, robustness and computation, with source code and a static replanning diagram linked.
Dual-Axis Solar Panel Tracker	• Designed a dual-axis panel mechanism in Fusion 360 and used light-dependent resistors to sense the direction of incident light. • Implemented microcontroller-based PID feedback for two servos and added an OLED display for live voltage and status monitoring. • Integrated the mechanical assembly, sensor readings and feedback loop in a physical instrumentation and control course prototype.

POSITIONS OF RESPONSIBILITY

Electronics Club **Core Team**	• Coordinated recruitment, equipment and inventory planning while supporting technical projects and workshops at IIT Madras. • Taught a hands-on parallel processing session at Shaastra and helped organize a hackathon on building an IoT smartwatch.
Electronics & Robotics Guild **Head**	• Led training in electronics and robotics, helped select members and assembled teams for Inter-IIT technical competitions. • Represented IIT Madras on the Bharat Forge swarm robotics challenge and coordinated resources for student engineering teams.

EXTRA-CURRICULAR

- Captained Alakananda Hostel to football titles in the Dean's and Schroeter tournaments and represented IIT Madras at intercollegiate sports festivals.
- Represented IIT Madras in BGMI as part of the institute's gaming contingent and placed first in the intra-IIT tournament during the 2023-24 season.